



NodeJS Introduction



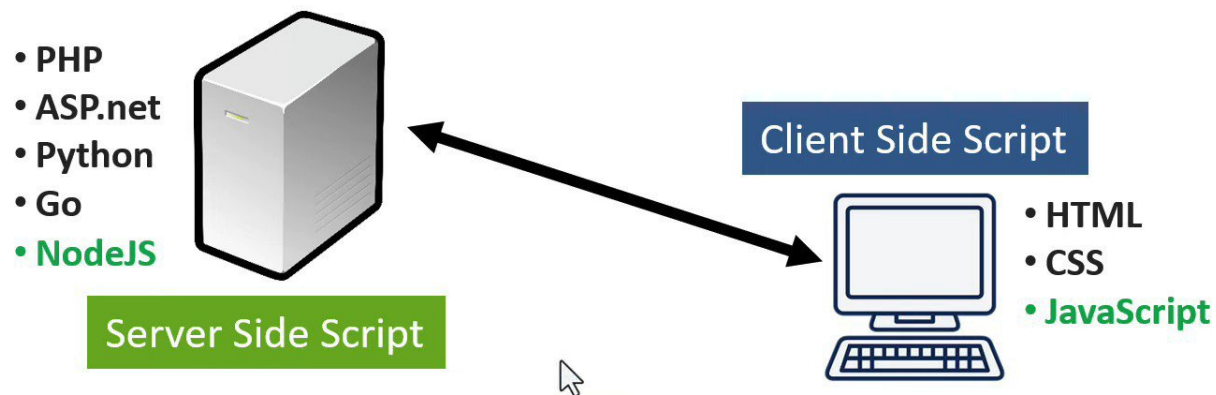
Node.js is an open-source, cross-platform **JavaScript runtime environment** that allows you to run JavaScript code outside a web browser.



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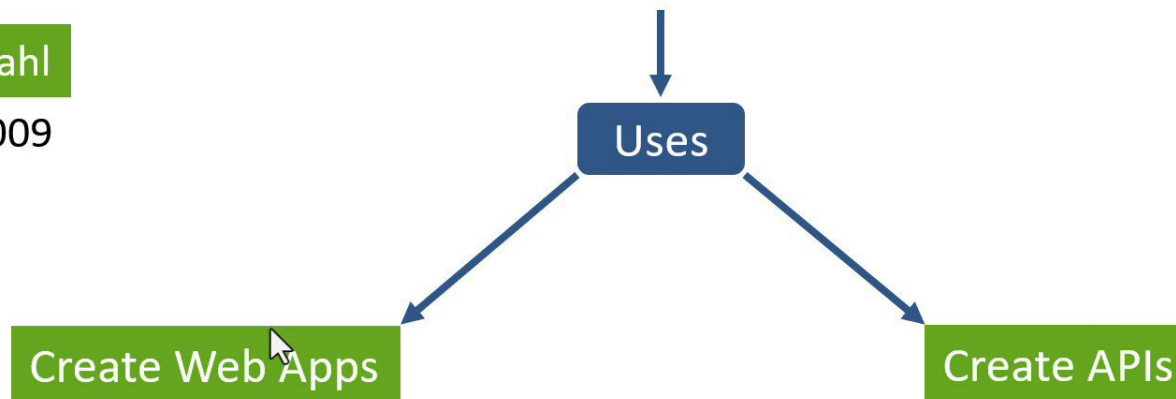
NodeJS Introduction



Node.js is an open-source, cross-platform **JavaScript runtime environment** that allows you to run JavaScript code outside a web browser.

Ryan Dahl

May 2009





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Chrome's V8 JavaScript engine

Uses

Create Web Apps

Create APIs

- Faster
- Lightweight
- Efficient
- Scalable



Use of NodeJS

- CMS (Content Management System)
- ERP(Enterprise Resource Planning)
- CRM(Customer Relationship Management)
- Mail server
- Real-time Applications : Chatting System, online games
- File Storage System
- Command-Line Tools
- Backend for Mobile Apps
- Streaming Data : Video streaming, Audio streaming, and processing sensor data from IoT devices.





Benefits of NodeJS

- **Performance:** V8 engine and non-blocking I/O make it very fast and efficient.
- **Scalability:** Easily handles a large number of concurrent connections.
- **Full-Stack JavaScript:** Developers can use JavaScript for both the front-end and back-end, which reduces context switching and improves development speed.
- **Large Ecosystem:** NPM provides a vast collection of libraries and tools.
- **Community:** A large and active community provides support, resources, and open-source contributions.
- **Real-time Capabilities:** Excellent for building real-time applications.
- **Easy to Learn:** If you already know JavaScript, the learning curve for Node.js is relatively gentle.
- **Cost-Effective:** Can often reduce infrastructure costs due to its efficient resource utilization.



Popular Companies using NodeJS

- Netflix
- LinkedIn
- Uber
- PayPal
- NASA
- Walmart
- Medium
- Trello
- eBay



ExpressJS Introduction



Express.js is a fast, minimal, and flexible **web framework** for **Node.js** that simplifies building **web applications** and **APIs** in a more structured and organized way.





What is web framework ?

Web frameworks are sets of **pre-written code** and **libraries** that provide a foundation for developing software applications.

Pre-written Code & Library

- Tools
- Components
- Modules





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Pre-written Code & Library

- Tools
- Components
- Modules

Examples

- Database Component
- Caching
- Pagination
- Session management
- Form Handling
- Security mechanisms
- User authentication
- APIs
- Payment Gateways



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Pre-written Code & Library

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Examples

- Database Component
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Benefits

- Code organization
- Reusability
- Standardization
- Testing & debugging support
- Community and support



Type of Frameworks

Opinionated Framework

Has a defined architecture (MVC, MVVM, etc.).

-
- Ruby on Rails (**Ruby**)
 - Django (**Python**)
 - Spring Boot (**Java**)
 - Laravel (**PHP**)

Unopinionated Framework

No strict rules or predefined workflow.

-
- Express.js (**Node.js**)
 - Flask (**Python**)
 - FastAPI (**Python**)
 - Koa.js (**Node.js**)
 - React.js



What we will learn in this series

- Parameters
- Requests
- APIs
- Template Engine
- Middleware
- Validation
- File Upload
- Routers
- Cookies
- Sessions
- Authentication
- Error Handling
- Testing
- Events
- Streams
- Sockets



VS Code Extensions

- ExpressJs 4 Snippets – ([Güray Yazar](#))
- ExpressJS Snippets – ([Xpertpk](#))
- MERN Snippets – ([mohamed-khaled](#))
- EJS language support – ([DigitalBrainstem](#))
- MongoDB for VS Code – ([MongoDB](#))
- Path Intellisense – ([Christian Kohler](#))
- ESLint – ([Microsoft](#))



NodeJS Project

npm init -y



Package.json

- Name
- Version
- Author
- Description
- Keywords





NodeJS Project

npm init -y

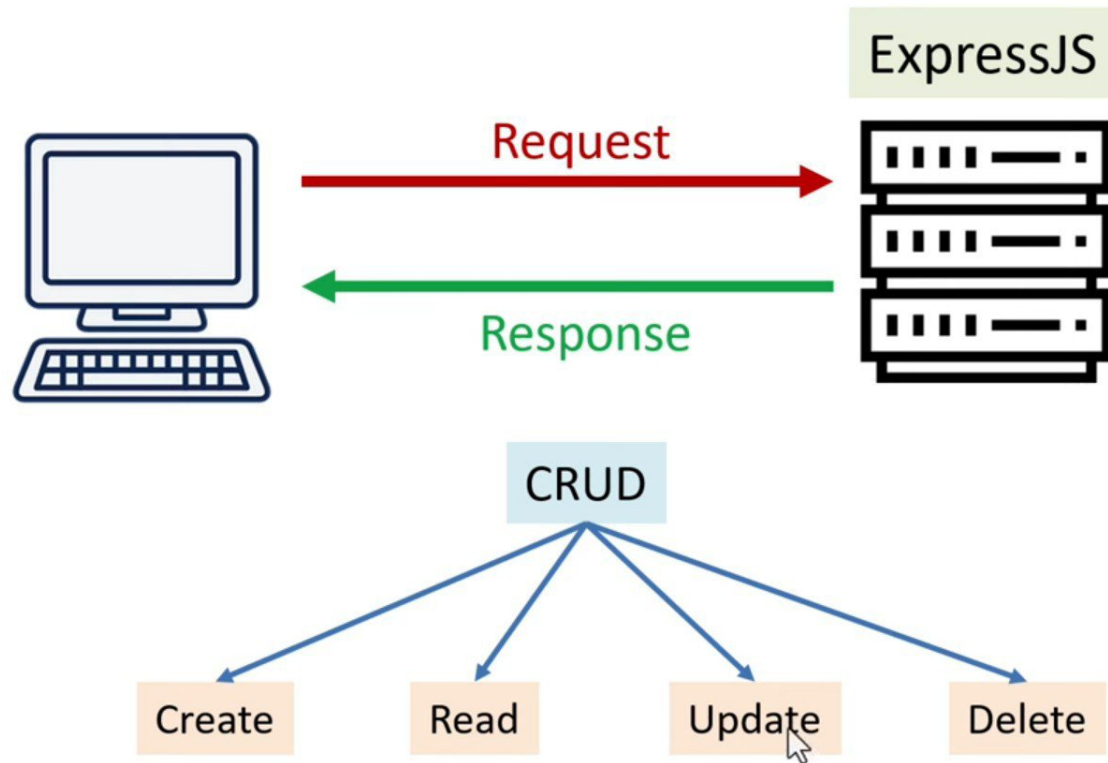


Package.json

- Name
- Version
- Author
- Description
- Keywords
- Packages (Dependencies)



How ExpressJS Works ?





How to use ExpressJS

1

```
const express = require('express')  
const app = express()
```

2

```
app.listen(3000, () => {  
  console.log(`Example app listening on port 3000`)  
})
```

HTTP requests

<http://localhost:3000/>

<http://localhost:3000/about>

<http://localhost:3000/gallery>



How to use ExpressJS

1 `const express = require('express')`
`const app = express()`

2 `app.listen(3000, () => {`
 `console.log(`Example app listening on port 3000`)`
`})`

HTTP requests

`http://localhost:3000/` →
`http://localhost:3000/about`
`http://localhost:3000/gallery`

```
app.get('/', (req, res) => {  
  res.send('Hello !')  
})
```



ExpressJS Route

Route



URL

- Webpage URL
- APIs URL

HTML

`http://localhost/about.html`

`http://localhost/pages/about.html`



ExpressJS Route

`http://localhost:3000/`

`http://localhost:3000/about`  `abc.html`

`http://localhost:3000/about/user` Nested Route

`http://localhost:3000/user/10` Route Parameters

`http://localhost:3000/user/yahubaba/rollno/12`

`http://localhost:3000/search?name=yahubaba`



ExpressJS Route

`http://localhost:3000/`

`http://localhost:3000/about`  `abc.html`

`http://localhost:3000/about/user` Nested Route

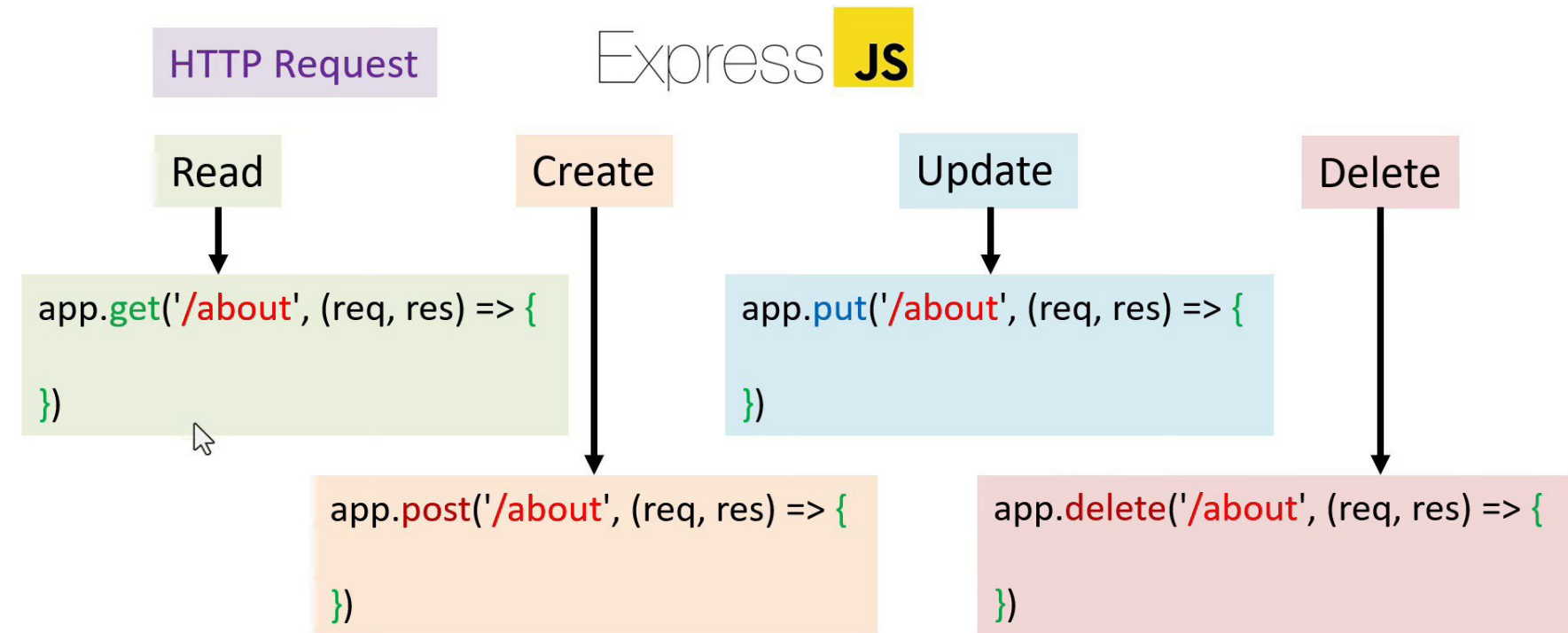
`http://localhost:3000/user/10` Route Parameters

`http://localhost:3000/user/yahubaba/rollno/12`

`http://localhost:3000/search?name=yahubaba` Query Parameters



When to use Route ?

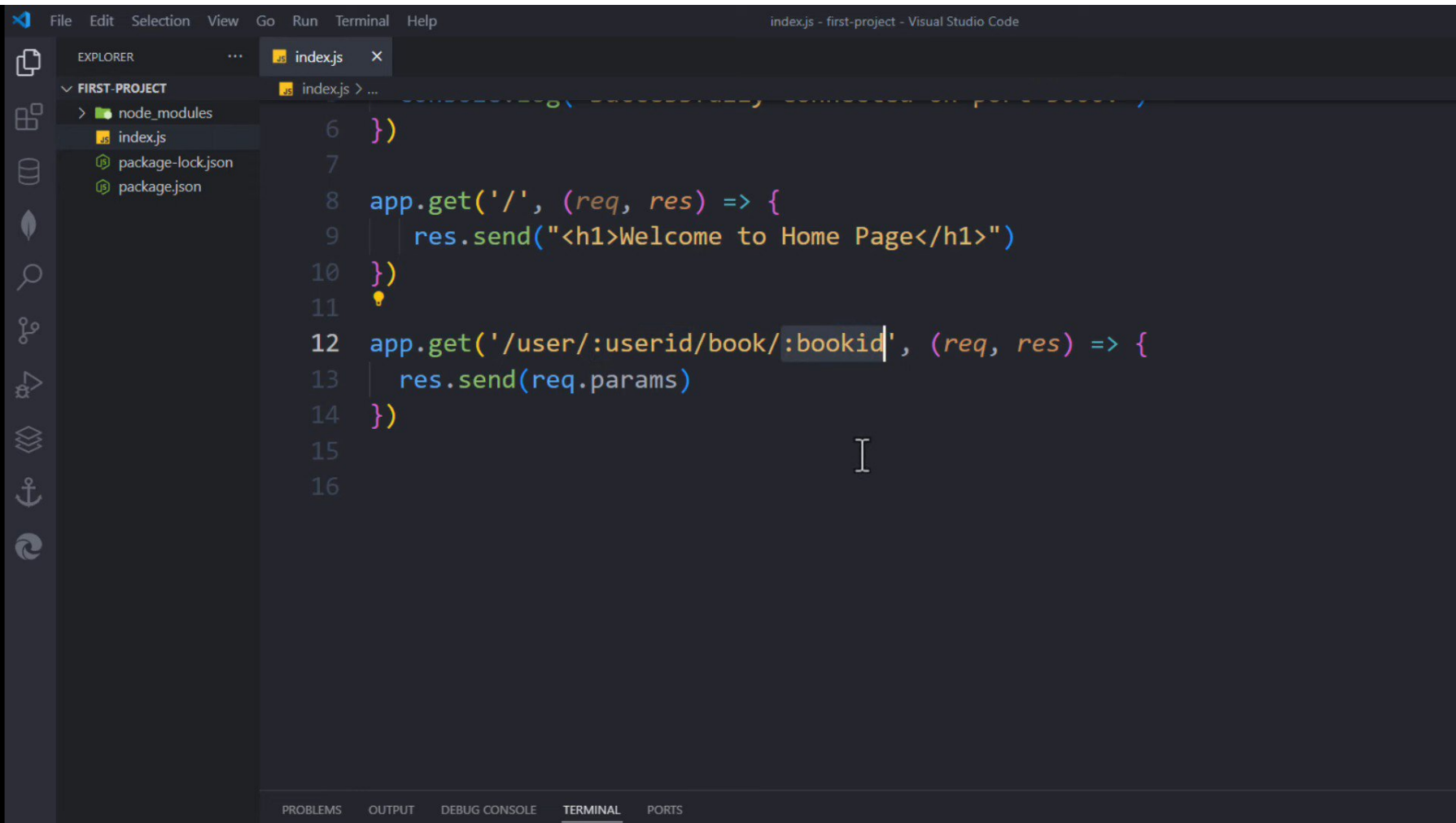


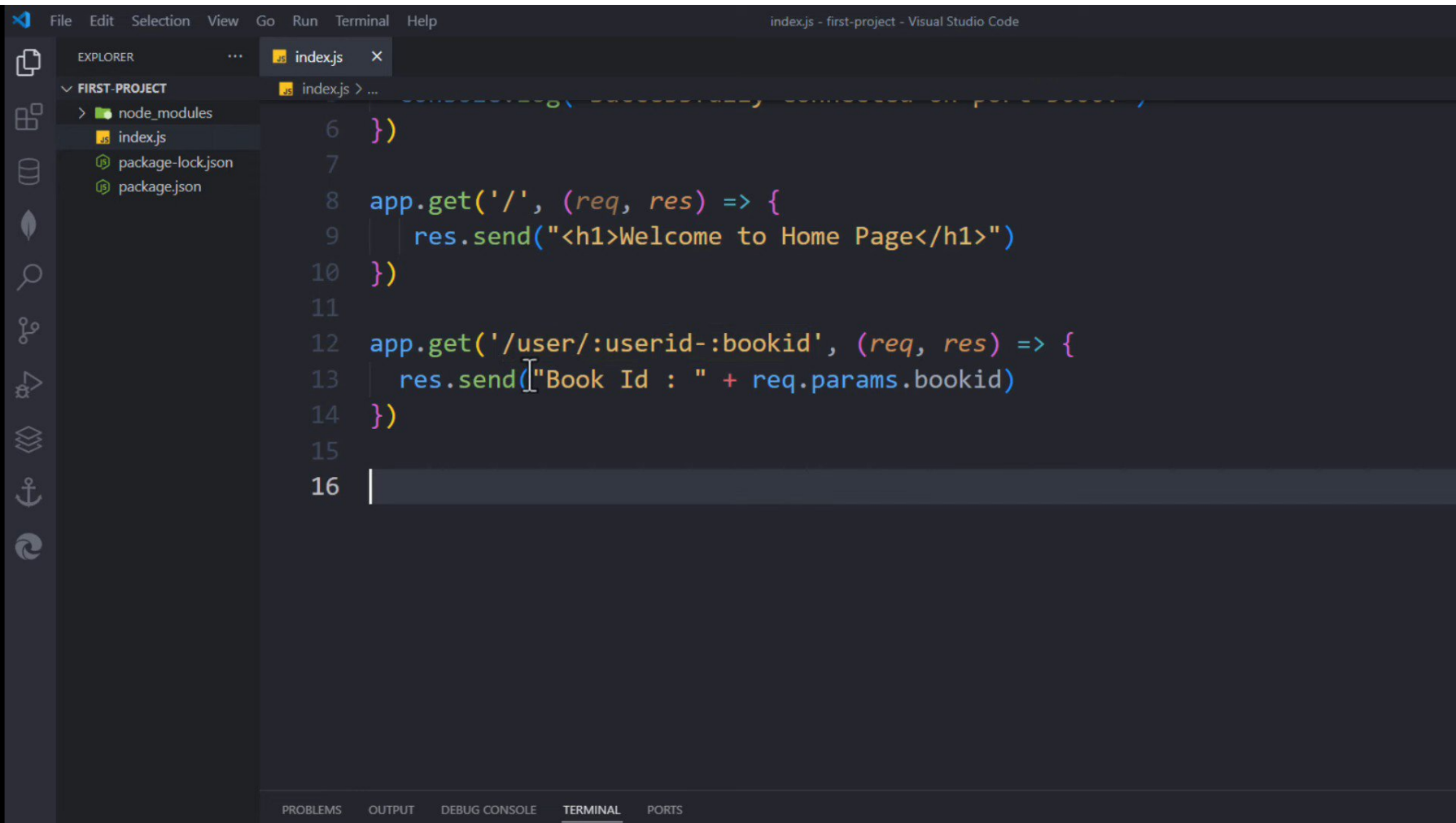


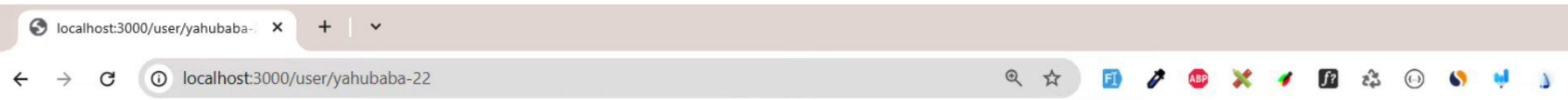
Pass Value through Route Parameters

http://localhost:3000/**user**/**10**

```
app.get('/user/:userid', (req, res) => {  
  res.send(req.params)  
})
```

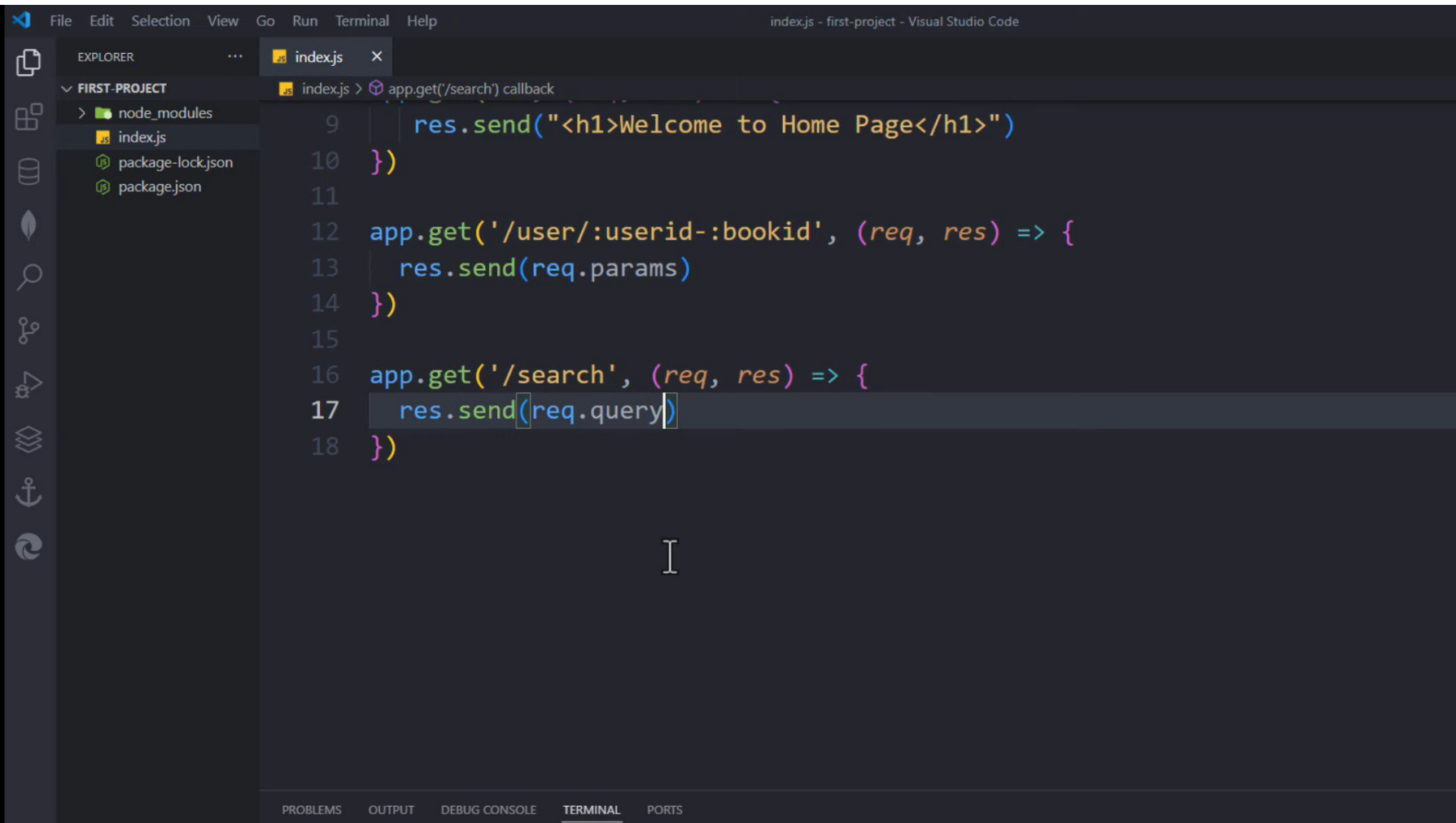







```
{  
  "userid": "yahubaba",  
  "bookid": "22"  
}
```

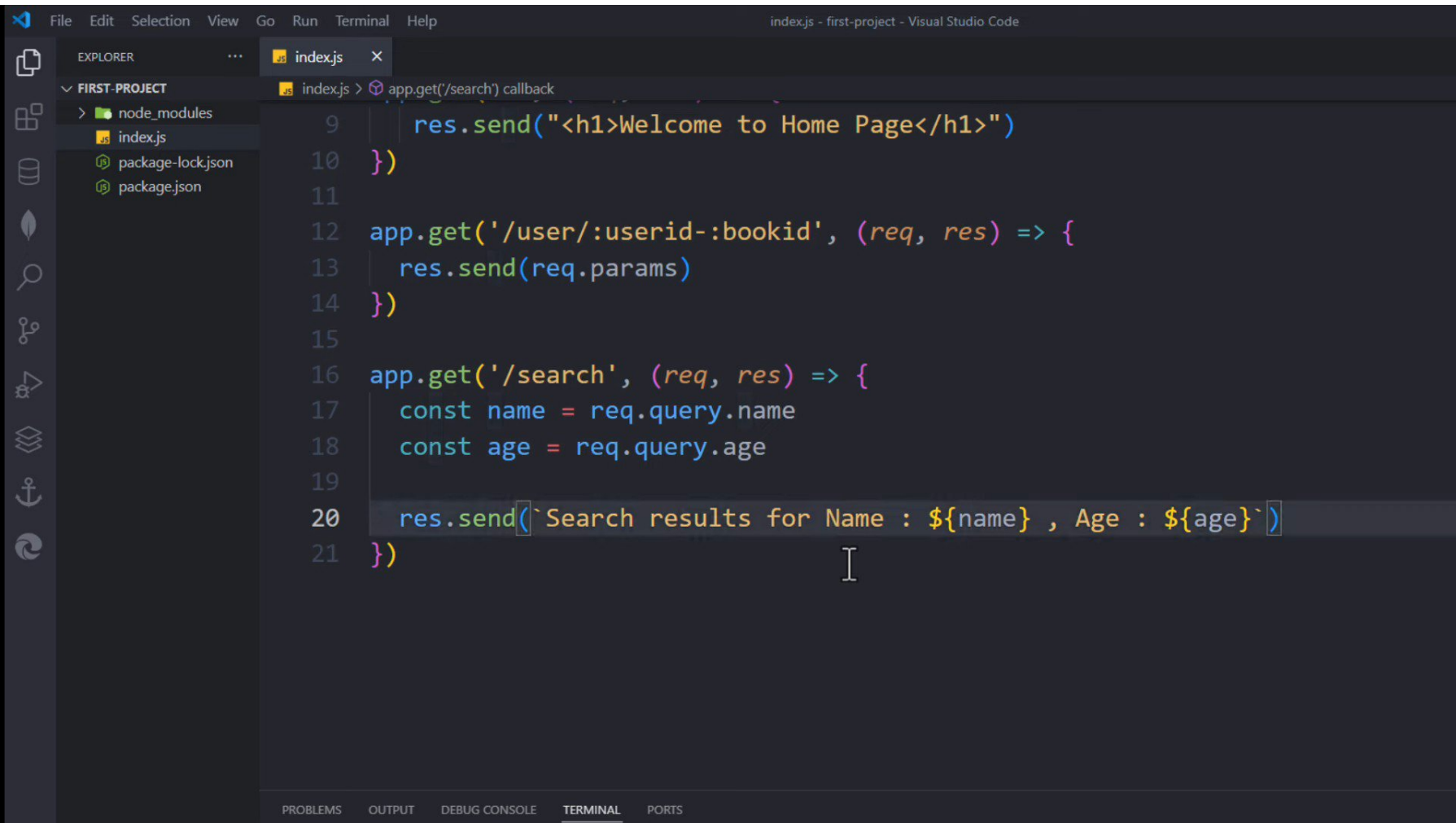
Raw

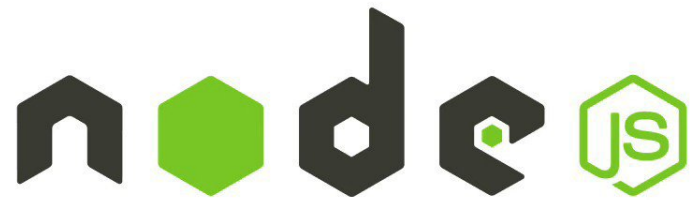


localhost:3000/search?name=y

localhost:3000/search?name=yahubaba&age=20&city=chandigarh

```
{  
  "name": "yahubaba",  
  "age": "20",  
  "city": "chandigarh"  
}
```





Response Methods

YahuBaba

www.Yahubaba.com



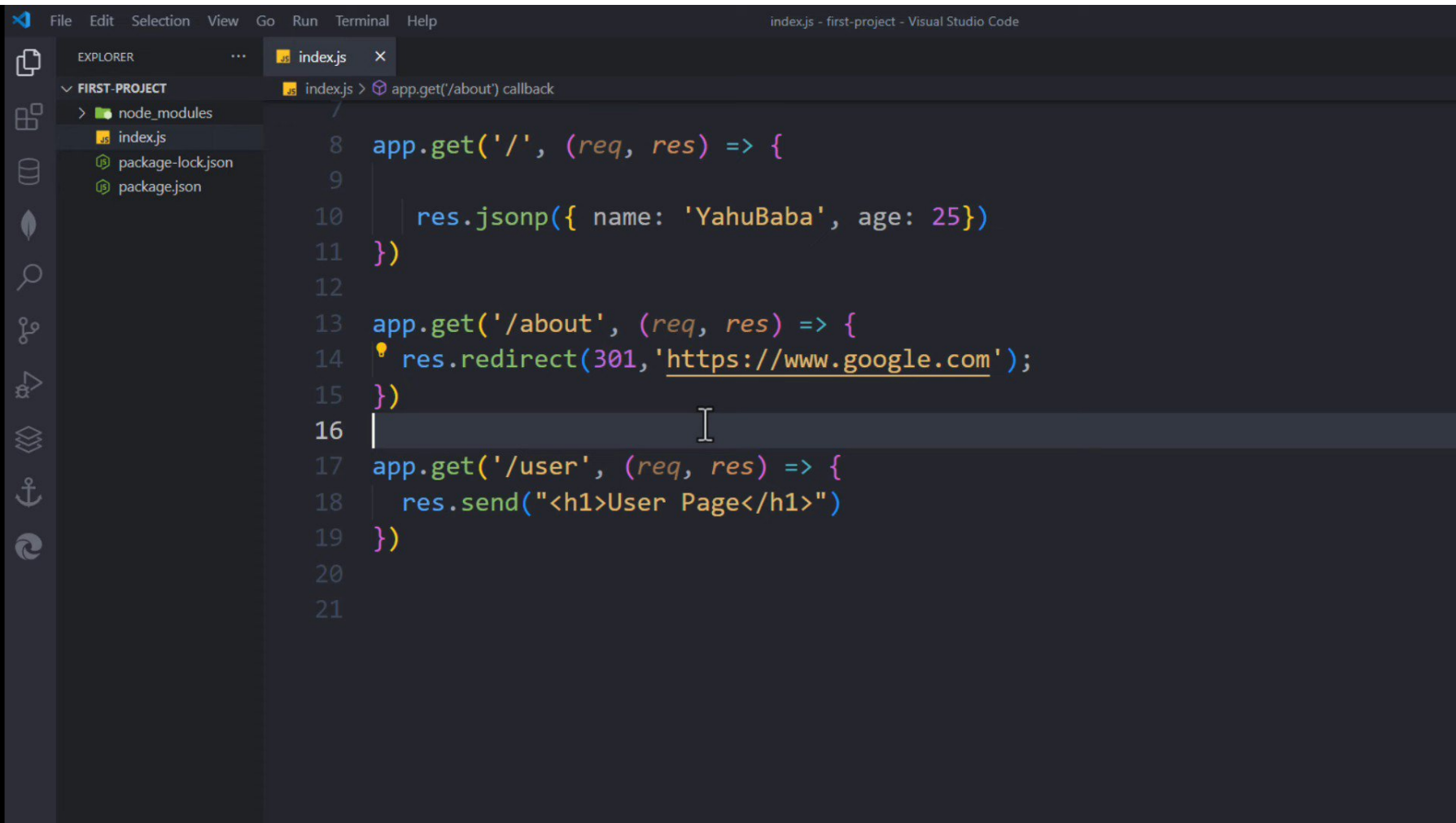
ExpressJS Response Methods

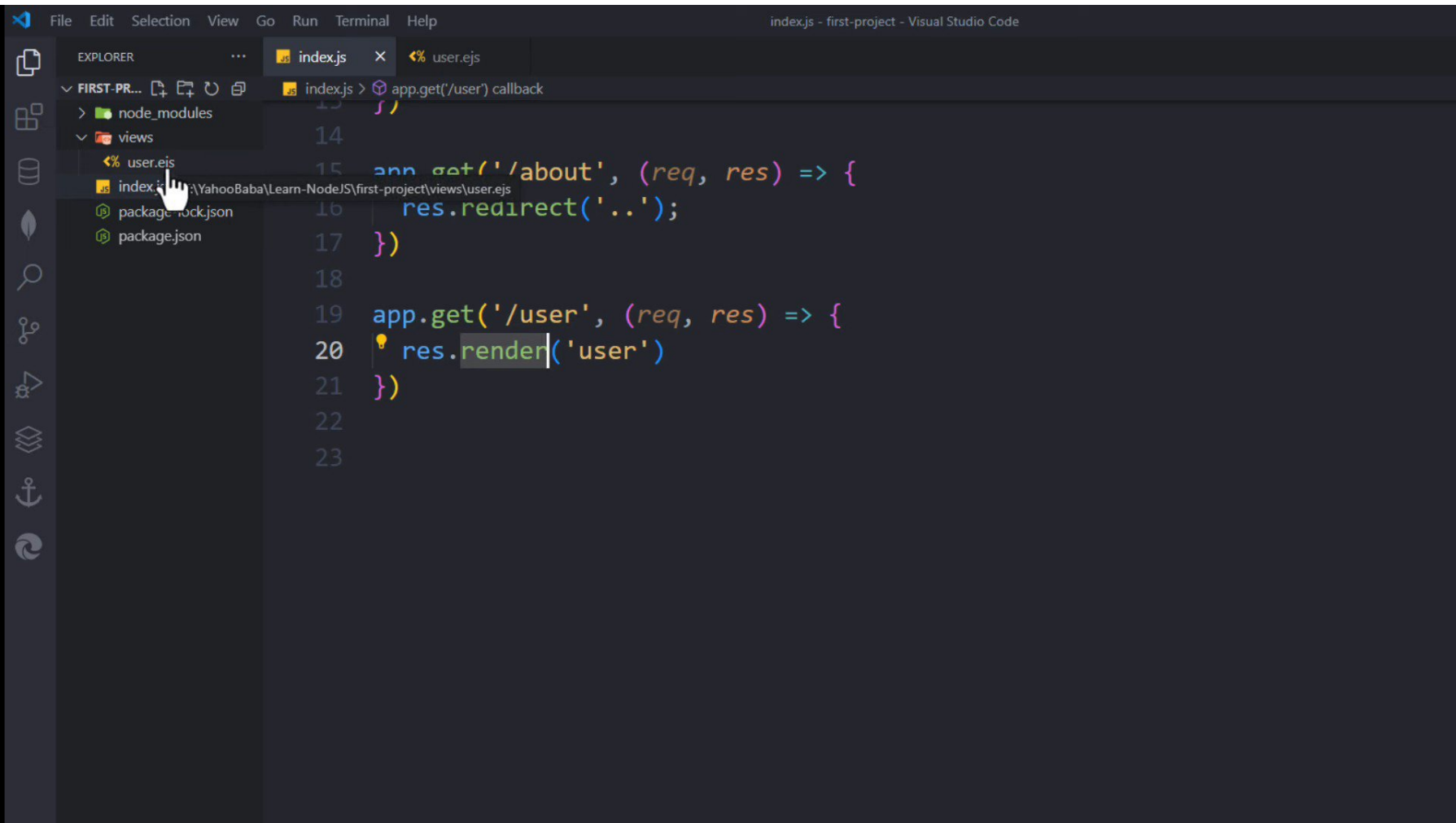
- `res.send()` → Text, HTML, Object, Array, Buffers
- `res.json()` → JSON (JavaScript Object Notation)
- `res.jsonp()` → JSONP (JSON with Padding)
- `res.redirect()`
- `res.render()` → Open HTML File using a template engine
- `res.download()` → PDF, Image File, Music, Video, Excel etc..
- `res.sendFile()`
- `res.end()`
- `res.sendStatus()`
- `res.headersSent()`
- `res.set()`
- `res.get()`

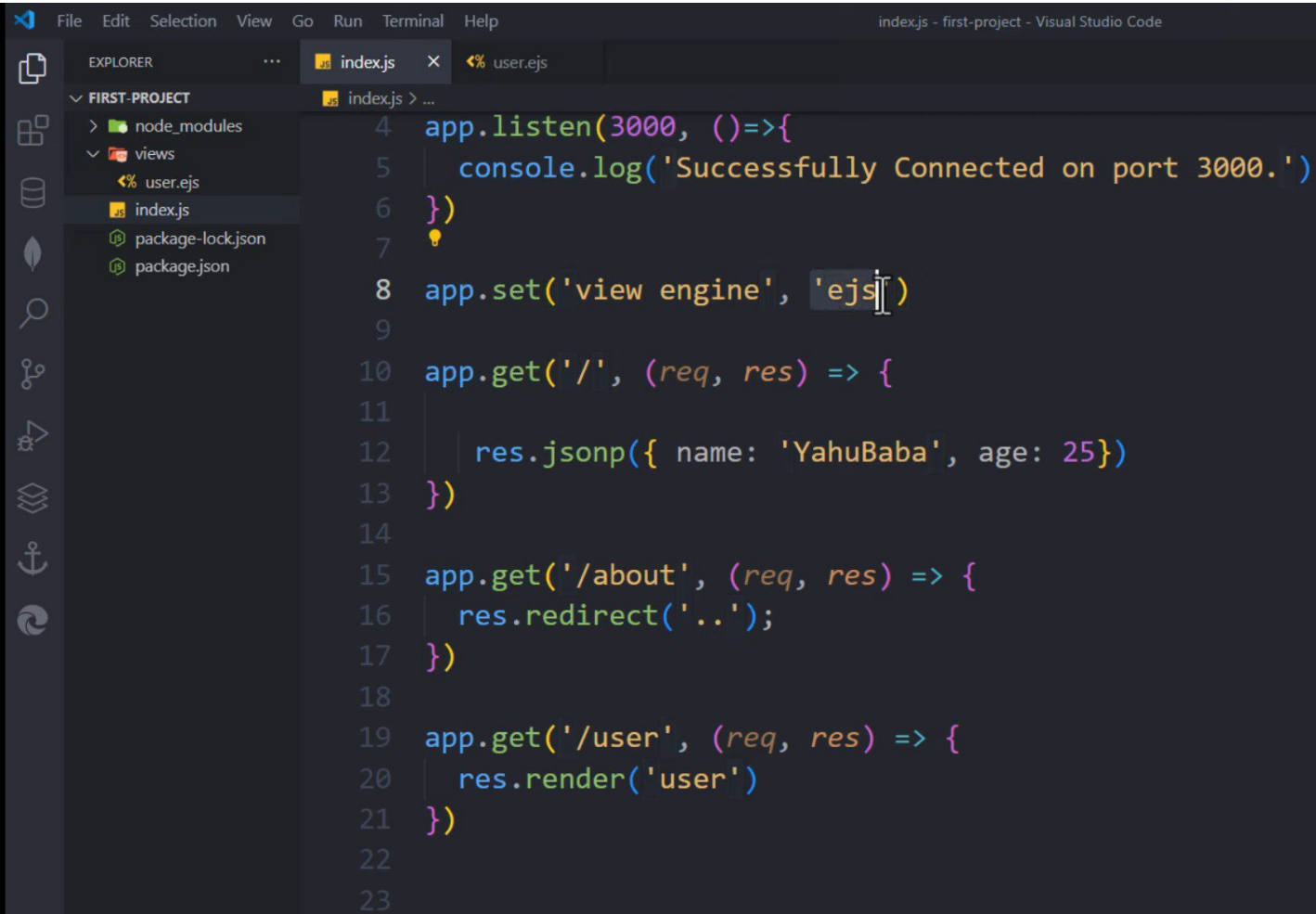


Redirect Code

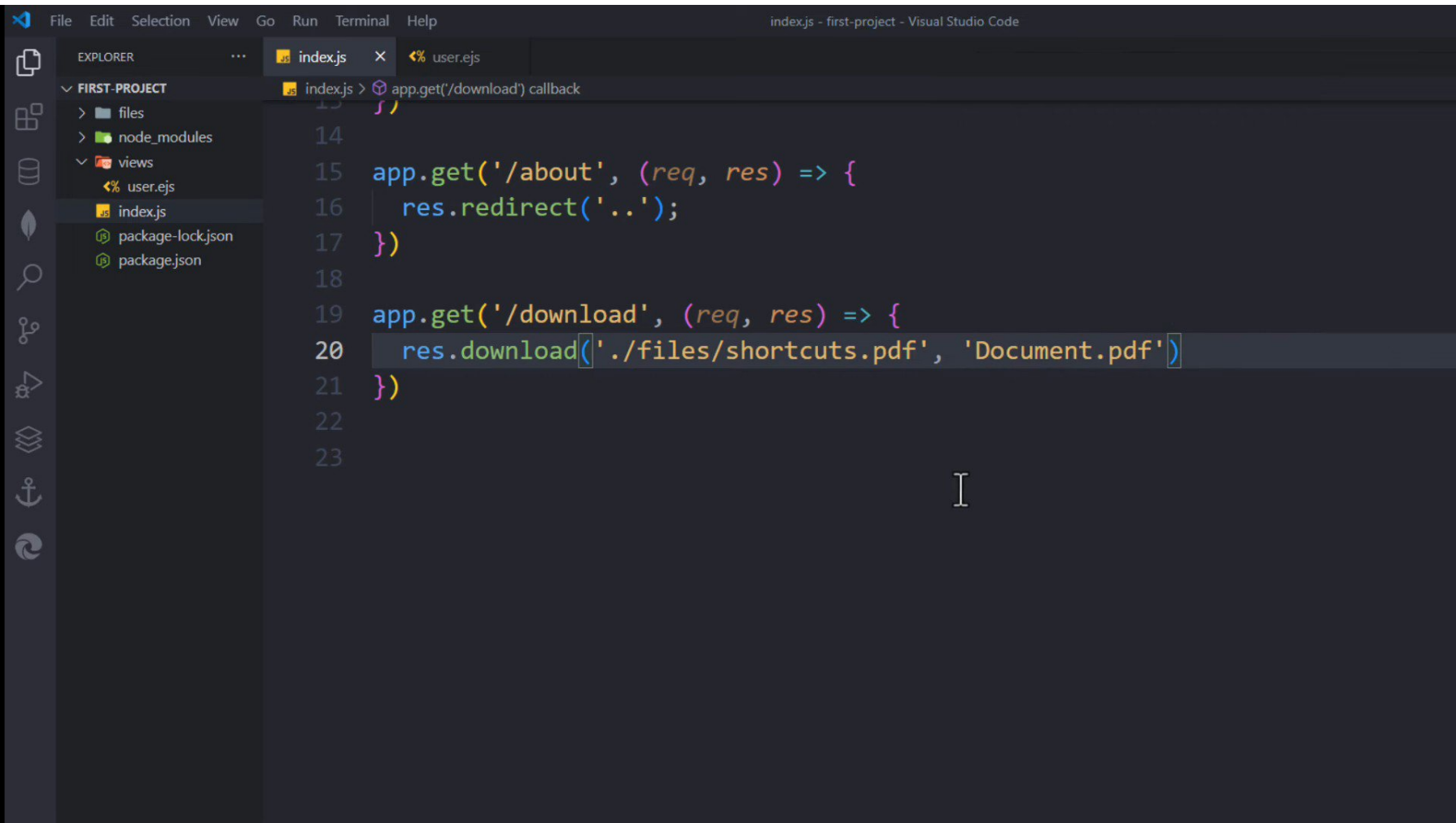
Code	Description
301	A permanent redirect used when a resource has been moved to a new URL.
302	A temporary redirect used when a resource has been temporarily moved to a new URL.
303	A temporary redirect used after a POST or PUT operation.
307	A temporary redirect similar to 302, but better for sites with non-GET operations.
308	The permanent counterpart to the 307 redirect.

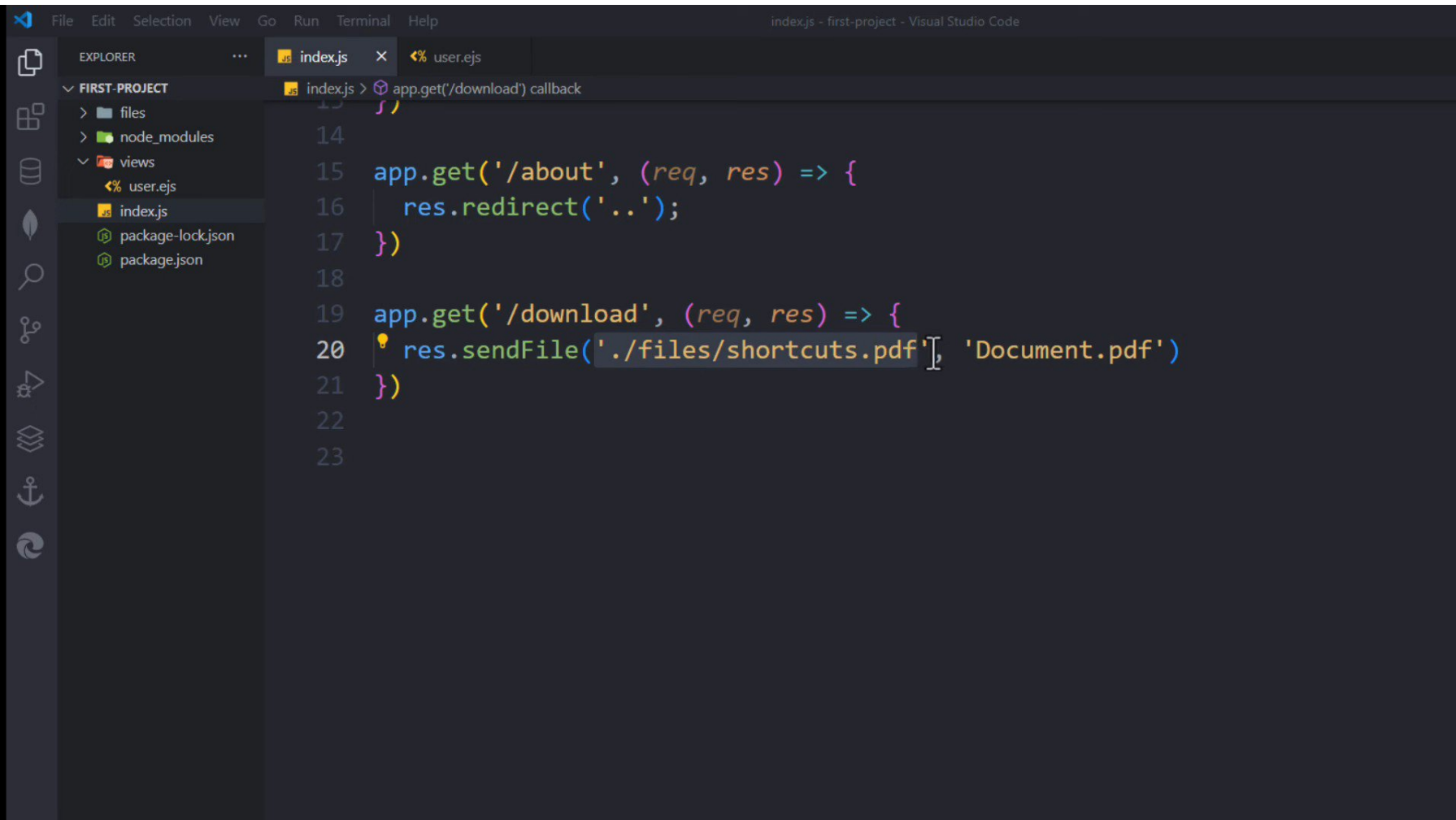


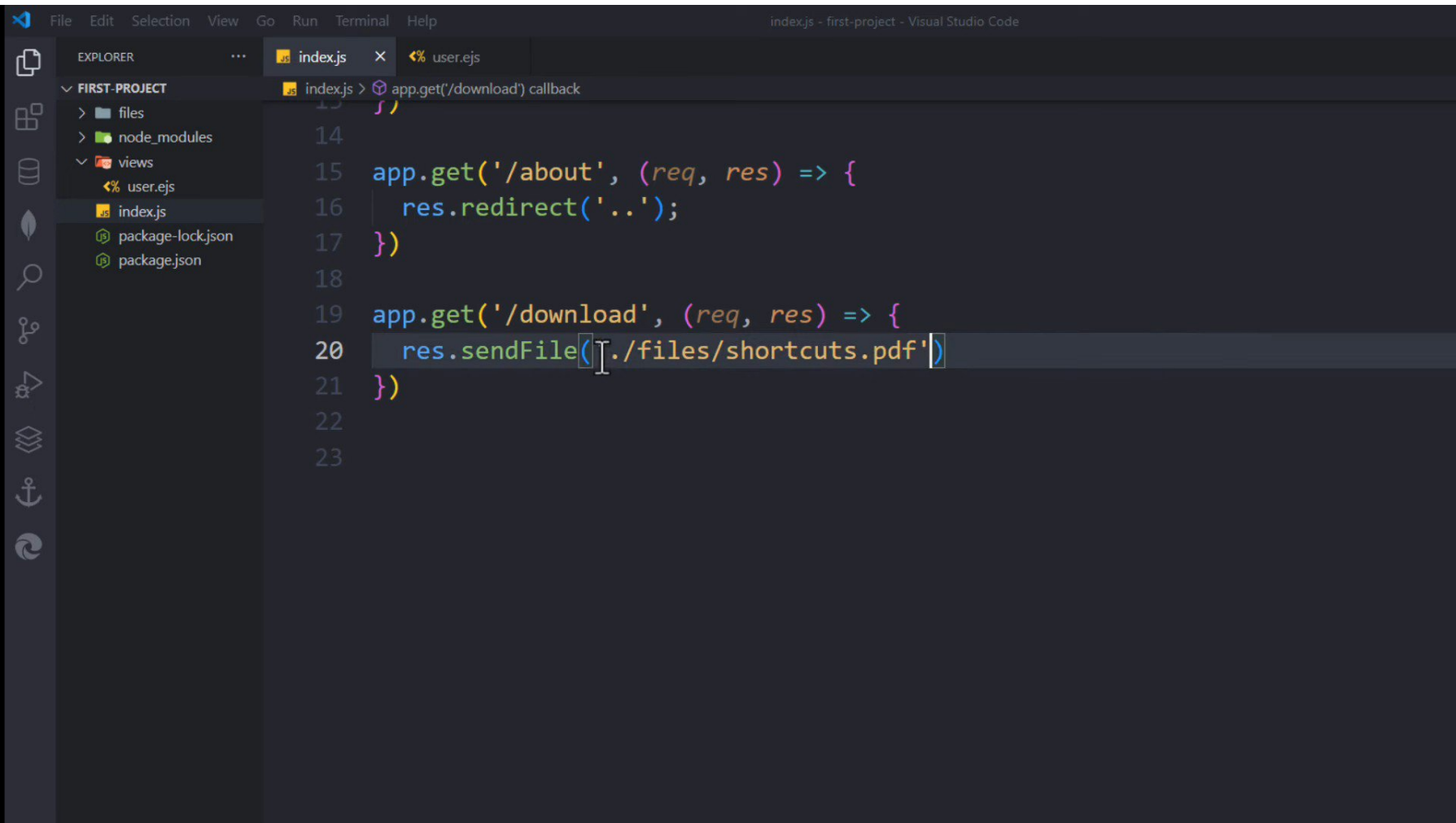


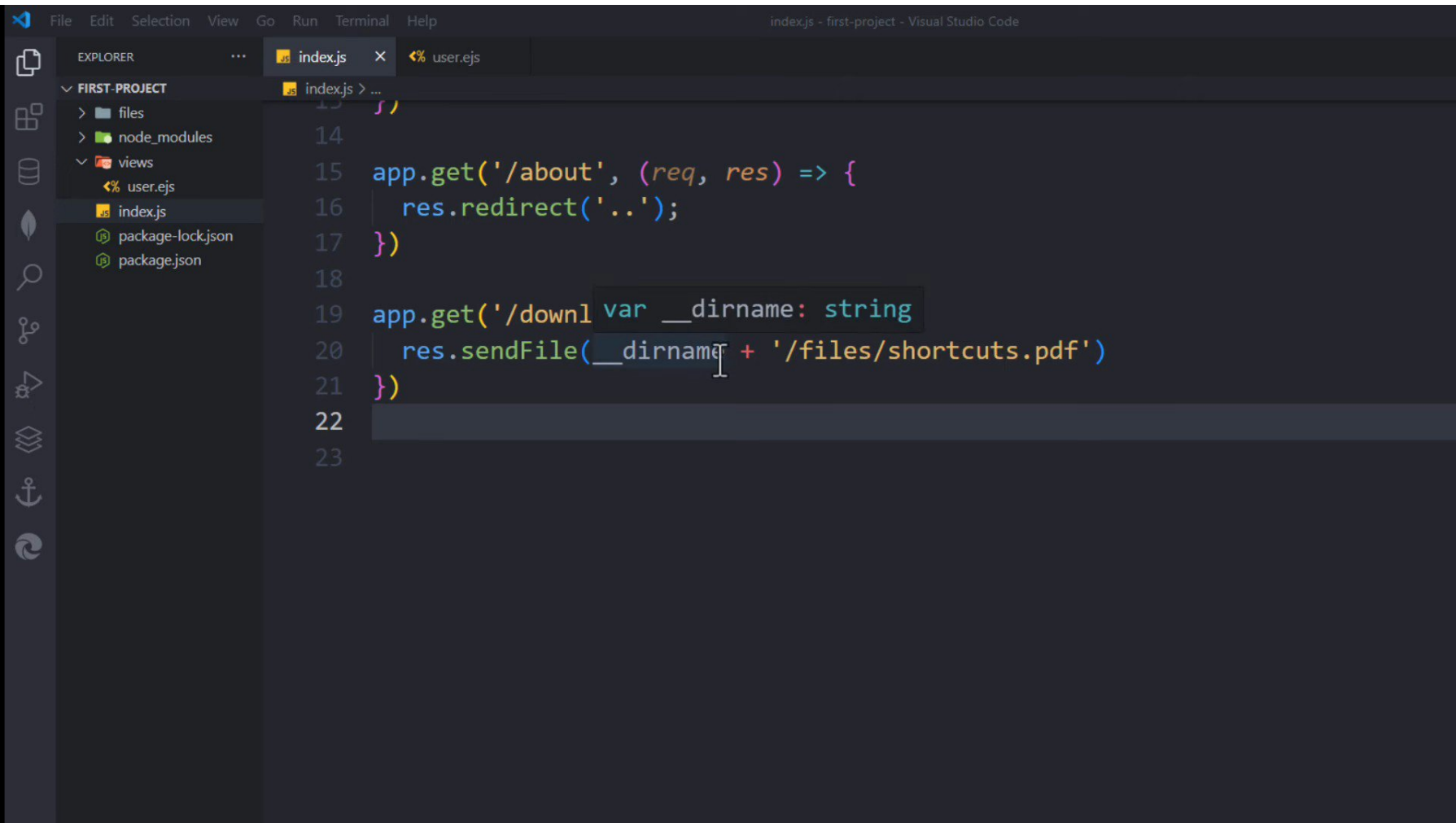


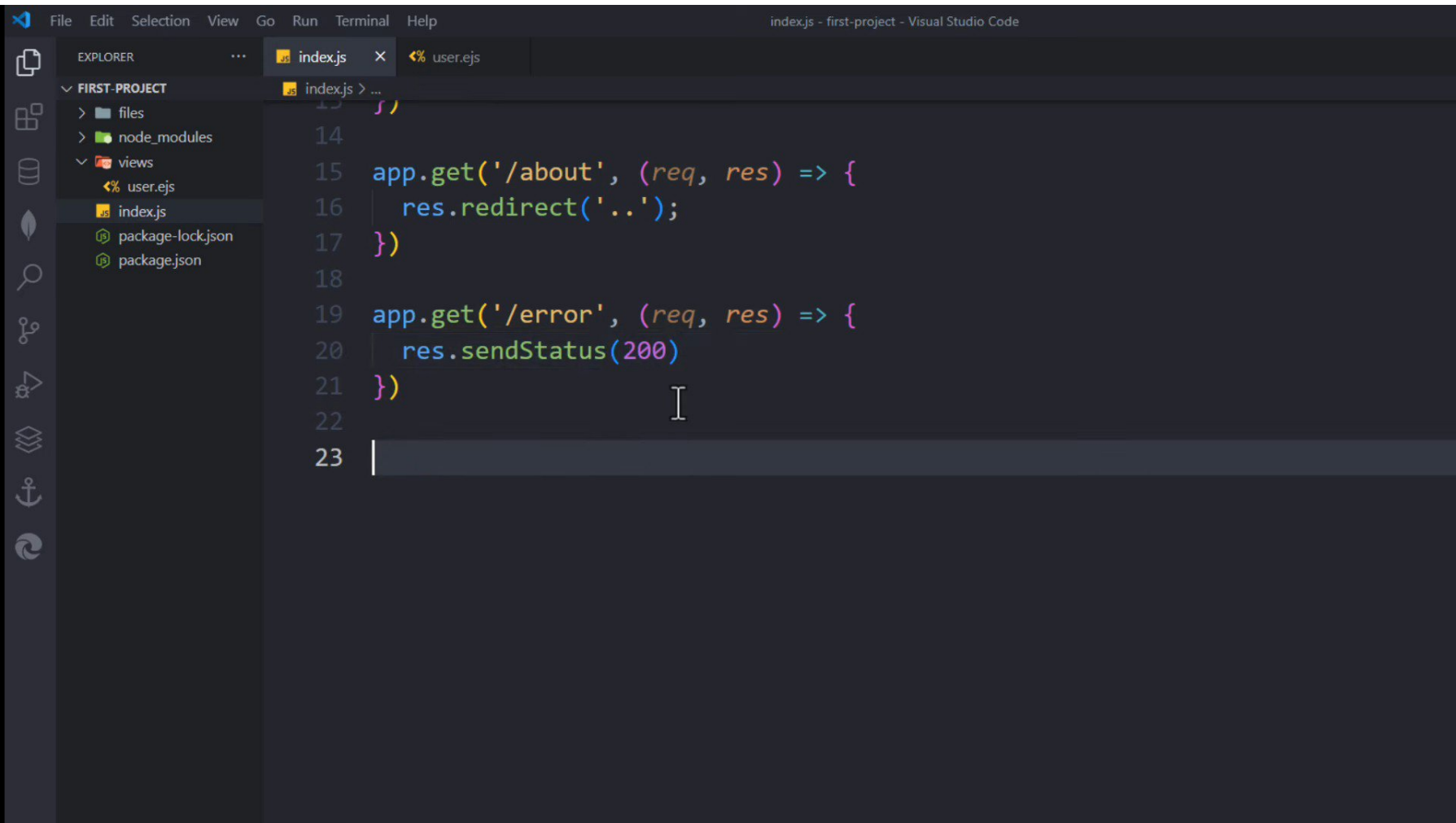
```
4 app.listen(3000, ()=>{
5   console.log('Successfully Connected on port 3000.')
6 })
7
8 app.set('view engine', 'ejs')
9
10 app.get('/', (req, res) => {
11   res.jsonp({ name: 'YahuBaba', age: 25})
12 })
13
14
15 app.get('/about', (req, res) => {
16   res.redirect('..');
17 })
18
19 app.get('/user', (req, res) => {
20   res.render('user')
21 })
22
23
```













Server Status Codes

200 - OK

201 - Created

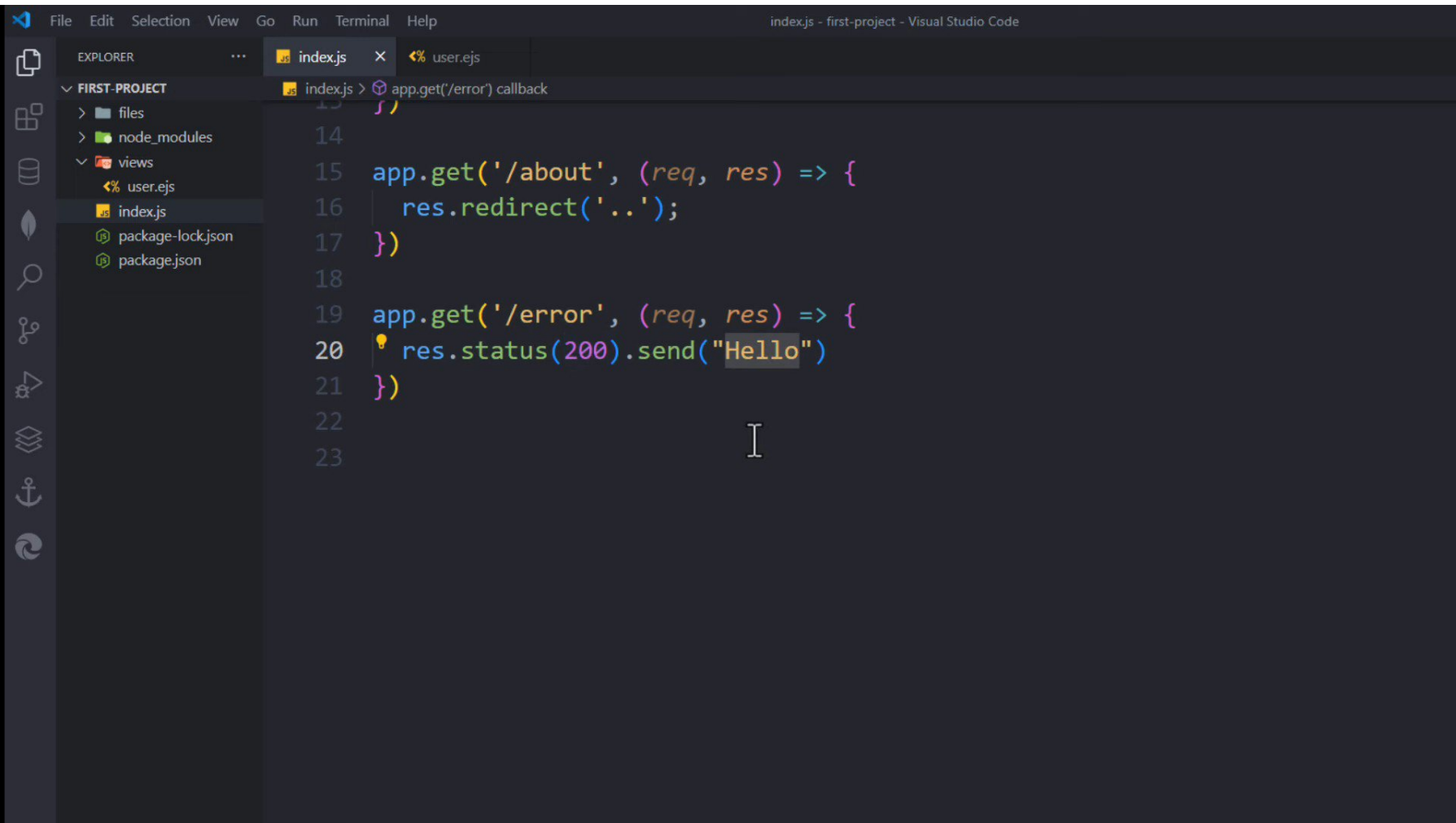
403 - Forbidden

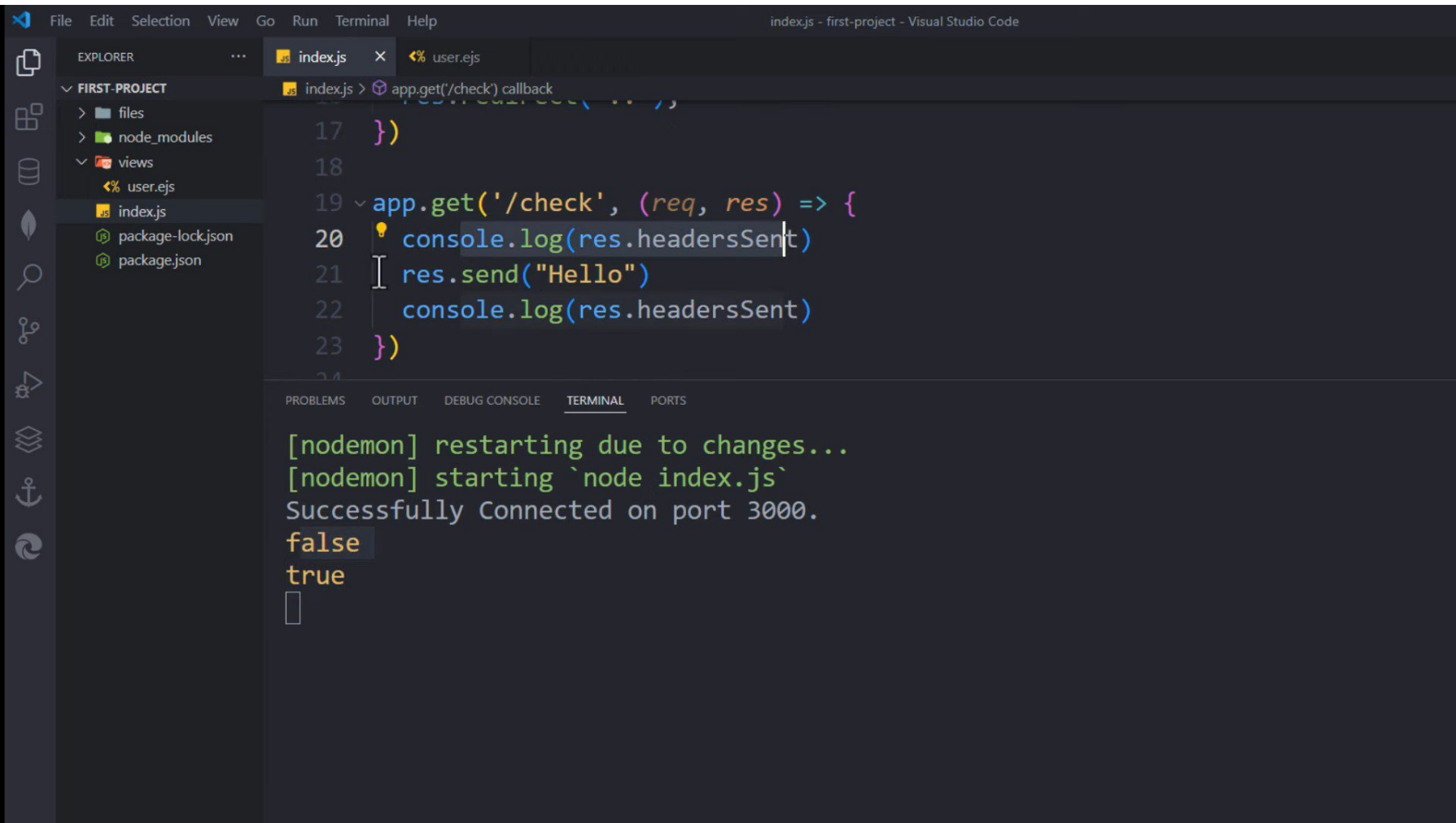
404 - Not Found

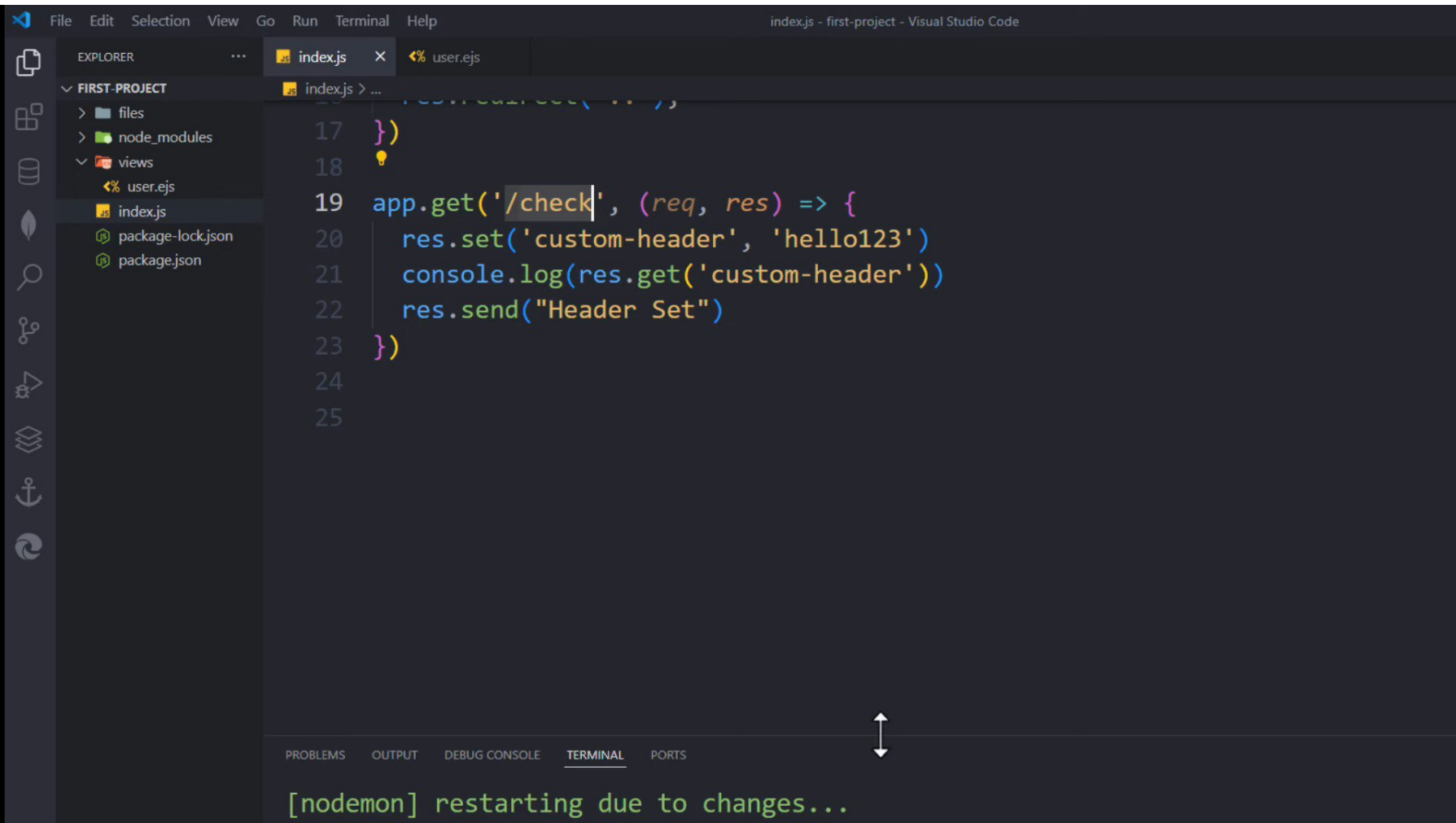
500 - Internal Server Error

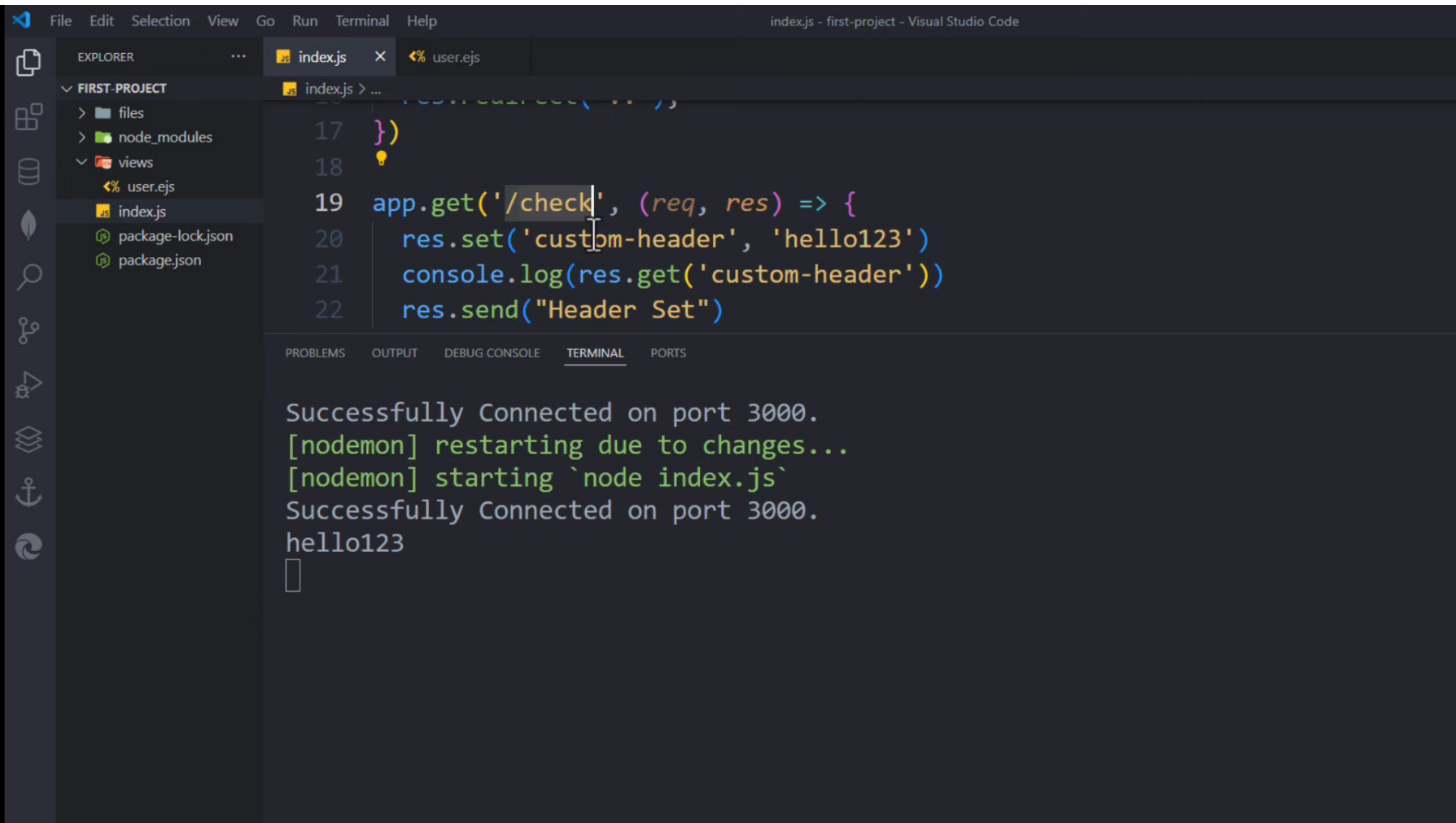
503 - Service Unavailable

504 - Gateway Timeout











Request Properties & Methods

YahuBaba

www.Yahubaba.com



ExpressJS Request Properties & Methods

Properties

- req.params
- req.query
- req.body
- req.cookies
- req.hostname
- req.ip
- req.ips
- req.method
- req.originalUrl
- req.path
- req.protocol
- req.secure
- req.route

JSON, form data

Methods

- req.accepts HTML, JSON, XML
- req.get
- req.is JSON, form data or plain text
- req.range




```
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2  const app = express()
3
4  app.listen(3000, ()=>{
5    console.log('Successfully Connected on port 3000.')
6  })
7
8  app.set('view engine', 'ejs')
9
10 app.use(express.json())
11
12 app.get('/', (req, res) => {
13   |
14   | res.jsonp({ name: 'YahuBaba', age: 25})
15   | })
16
17 app.get('/about', (req, res) => {
18   |
19   | })
20
21
```

